

Project Proposal Guide (Weeks 7–8)

This worksheet guides the team in developing the Capstone/Software Engineering Project Proposal. It integrates the refined problem statement, findings from the Review of Related Literature (RRL), and the proposed solution concept.

Team Information

Project Title: **ResQMesh: A Decentralized Emergency Messaging System Using Bluetooth and WiFi Direct**

Project Short Description (<20 words): Offline disaster messaging system using device-to-device data hopping without cellular or internet connectivity.

Team Code: 2526-sem2-cs342-16

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PART 1: Introduction (Approx. 300 words)

Natural disasters such as typhoons and floods frequently disrupt communication infrastructure, especially in vulnerable regions like the Philippines. During severe weather events, cellular towers often fail due to power outages, physical damage, or network congestion [17]. This results in the inability of affected individuals to send emergency messages, request rescue, or coordinate with responders. Studies in disaster communication emphasize that communication failure significantly delays rescue operations and increases casualty risks [18].

Existing communication systems rely heavily on centralized infrastructure, where messages are transmitted from sender to telecom towers and then routed to recipients. However, this model becomes ineffective when infrastructure collapses [7]. While some emergency systems exist, many still depend on internet connectivity or pre-established networks, limiting their usability in fully offline environments [8].

Recent advancements in mobile device capabilities such as Bluetooth and WiFi Direct enable direct device-to-device communication within short ranges [11][12]. Prior studies on delay-tolerant networks (DTNs) and opportunistic networking show that data can propagate through intermediate devices using a store-carry-forward mechanism [2], [4]. However, current implementations are limited by usability issues, lack of structured message routing, and absence of scalable adoption strategies for real-world disaster scenarios [1][6].

Furthermore, the inability to pinpoint survivor locations without internet-dependent mapping services severely hampers targeted rescue operations, as traditional GPS tracking and map rendering fail when cellular data is unavailable [21][22].

The identified research gap lies in the absence of a user-friendly, scalable, and infrastructure-independent emergency messaging system that leverages ubiquitous mobile devices for reliable communication and spatial awareness during disasters.

This project proposes ResQMesh, a decentralized messaging system that enables emergency communication through data hopping across nearby devices, ensuring message delivery and offline survivor localization even without cellular or internet connectivity.

PART 2: Objectives

General Objectives (System Modules)

G01: Offline Message Transmission Module

Automate direct offline message delivery between nearby mobile devices without dependence on cellular or internet connectivity by achieving $\geq 90\%$ successful message transmission within a 20-meter Bluetooth/WiFi Direct range.

Specific objectives for G01:

- 1.1 Automate nearby device detection to establish peer-to-peer connections in ≤ 15 seconds.
- 1.2 Optimize device-to-device message transmission (Device A to Device B) with ≤ 5 seconds latency within Bluetooth/WiFi Direct range.

G02: Network Relay and Broadcasting (Data Hopping) Module

Increase message reachability by extending transmission coverage through multi-device relay within 5 minutes.

Specific objectives for G02:

- 2.1 Automate message forwarding to nearby devices with $\geq 85\%$ relay success rate
- 2.2 Maintain message propagation across a maximum of 5 intermediate devices to ensure network stability (multi-hop)
- 2.3 Improve public broadcasting efficiency to achieve a $\geq 75\%$ successful local network saturation rate.

G03: Privacy and Security Module

Enhance communication security by enforcing 100% end-to-end encryption to eliminate unauthorized message exposure.

Specific objectives for G03:

- 3.1 Automate the encryption of private messages to achieve a 100% secure transmission rate.
- 3.2 Reduce unauthorized access risks by maintaining a 0% decryption success rate for unintended recipient devices.

3.3 Automate the assignment of unique hashed identifiers to guarantee zero user duplication across the network.

G04: Message Routing and Relay Verification Module

Improve indirect message delivery by optimizing message routing and relay verification to achieve $\geq 85\%$ successful multi-hop transmission accuracy.

Specific objectives for G04:

4.1 Optimize relay path selection to reduce redundant message forwarding to $\leq 10\%$ of total network traffic.

4.2 Reduce broadcast storms by ensuring previously relayed messages are echoed 0 times by the same device.

4.3 Automate delivery status verification to provide senders with $\geq 90\%$ confirmation accuracy for indirect recipients.

Research Questions

1. How effectively can device-to-device communication maintain message transmission without network infrastructure?
2. To what extent can data hopping improve communication coverage during disasters?
3. How secure and private is the transmission of messages in a decentralized environment?
4. How quickly can emergency broadcasts reach nearby users using opportunistic networking?

PART 3: Methods

The proposed system is a **mobile-based decentralized communication platform** that enables users to send and receive messages through Bluetooth and WiFi Direct. Messages are transmitted using a **store-carry-forward mechanism**, where nearby devices act as relays to propagate messages across a network of users. The system targets civilians, disaster victims, and emergency responders.

The development will follow the **Agile methodology**, allowing iterative development of system modules such as messaging, relay, and security. Each sprint will focus on implementing and refining a specific module, ensuring flexibility and continuous improvement based on testing feedback.

System validation will involve **performance testing and simulation scenarios**. Metrics include:

- Message delivery success rate (%)
- Transmission latency (seconds)
- Relay success rate (%)
- Coverage expansion (% increase in reach)

These metrics will demonstrate how effectively the system improves communication reliability during network outages.

PART 4: Expected System

Key Features (MVP)

- Offline messaging via Bluetooth/WiFi Direct
- Public (broadcast) and private messaging
- Automatic message relay (data hopping)
- Unique user identifier system
- Message prioritization for emergencies
- Encrypted communication for privacy

System Workflow

1. User composes a message (public or private)
2. Message is broadcast to nearby devices within range
3. Receiving devices store and forward the message to other nearby devices
4. Process continues until:
 - Message reaches the intended recipient (private), OR
 - Message spreads across the network (public)
5. Recipient device decrypts and displays the message

PART 5: Discussion

Scope

The system focuses on short-range communication (50–100 meters per device) and multi-hop propagation using Android devices. It operates fully offline without requiring internet or cellular networks.

Limitations

- Requires sufficient device density for effective propagation
- Battery consumption may increase due to continuous scanning
- Limited to Bluetooth/WiFi Direct range per hop
- Adoption depends on user participation

Expected Contribution

ResQMesh introduces a practical, infrastructure-independent communication system that enhances disaster response capabilities. By combining decentralized data-hopping with offline spatial awareness, it addresses the research gap by providing a scalable solution that improves both communication accessibility and targeted survivor localization during emergencies.

PART 6: Traceability Matrix

RRL Finding/Theme	Identified Gap	Research Question	Proposed Function
Disaster communication failure	Dependence on cellular networks	How to communicate without infrastructure?	Offline messaging module
Opportunistic networking	Limited real-world usability	How effective is data hopping?	Relay system

Mobile device connectivity	Lack of scalable adoption	Can smartphones act as nodes?	Bluetooth/WiFi Direct communication
Security in decentralized systems	Weak privacy protection	How secure is message transmission?	Encryption module
Emergency alert systems	Slow broadcast delivery	How fast can alerts spread?	Public broadcast feature

PART 7: References

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